

Robust and Expressive Humanoid Motion Retargeting via Optimization-Based Rig Unification

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Abstract—Humanoid robots are increasingly being developed for seamless interaction with humans in diverse domains, yet generating expressive and physically-feasible motions remains a core challenge. We propose a robust and automated pipeline for motion retargeting that enables the generation of natural, stable, and highly expressive motions for a wide variety of humanoid robots using different motion data sources, including noisy pose estimations. To ensure robustness, our approach unifies motions from different kinematic structures into a common canonical rig, systematically refines the motion trajectory to address infeasible poses, enforces foot-contact constraints, and enhances stability. The retargeted motion is then refined to closely follow the source motion while respecting each robot’s physical limits. Through extensive experiments on 12 simulated robots and validation on three real robots, we show that our methodology reliably produces expressive upper-body movements with consistent foot contact. This work represents an important step towards automating robust and expressive motion generation for humanoid robots, enabling deployment in various real-world scenarios.

I. INTRODUCTION

Humanoid robots have advanced rapidly in recent years, with numerous research institutions and companies developing increasingly sophisticated platforms [1], [2]. These robots are designed to interact with humans and perform complex tasks in a number of domains such as industry, entertainment, and education. As they become more prevalent in daily life, the demand for highly expressive and natural robot motions continues to grow [3].

However, generating expressive yet physically feasible motions remains challenging due to the need to balance robot-specific constraints (e.g., joint limits, balance, and mass distribution) with aesthetic considerations [4]. Manual motion design is time-consuming and labor-intensive, highlighting the need for methods that can automatically generate feasible, life-like motions for humanoid robots. Specifically, our approach is particularly suited for application domains

such as telepresence, entertainment robots, and interactive robots, where expressive upper-body motions with stable, stationary foot placement are critical.

In particular, our approach emphasizes robustness to noisy or incomplete source data, such as 3D pose estimation from RGB videos, enabling the generation of a broader range of robot motions. This often requires correcting large inaccuracies in body orientation or foot placement while still preserving the overall expressiveness of the source motion.

In many scenarios (e.g., industrial tasks or static performances), the robot’s feet often remain in full contact with the ground to maintain a stable center of mass (COM). Thus, we primarily focus on generating precise and expressive upper-body motions while keeping the robot’s feet stationary, rather than dynamic stepping or locomotion.

In this paper, we propose an automated robust robot motion retargeting method (Figure 1) that strongly emphasizes robustness and expressiveness in foot-fixed scenarios. Our method addresses noisy or infeasible source motions, whether they come from motion capture data [5], [6] or 3D pose estimation methods [7]–[9]. In particular, we propose a two-step motion refinement process that (i) detects and corrects large errors in foot placement and center-of-mass alignment, and (ii) preserves the temporal style of the original motion while enforcing physical feasibility.

We unify diverse motion data into a single canonical rig (which will be called a common-rigging process), systematically correct foot positioning and self-collisions, then map the refined motion to each robot using a direction vector-based retargeting method in combination with inverse kinematics. Our key contributions are:

- 1) **A common-rigging pipeline** that unifies and refines source motion data (whether from MoCap or noisy 3D pose estimations) into a physically consistent human-like rig, correcting foot placement, upright torso, and self-collisions.
- 2) **A generalizable motion retargeting approach** that robustly adapts the refined motion to various humanoid robots through semantic correspondence, using trajectory optimization to ensure physical feasibility under joint and collision constraints.
- 3) **Extensive experimental validation** on 12 simulated robots and on three real hardware platforms, demonstrating that our approach produces stable, expressive, and physically plausible motions even from noisy input data.

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Fig. 1: Retargeting human motions from RGB videos onto physical robot platforms, (A) AMBIDEX, (B) THORMANG, and (C) JF2.

II. RELATED WORK

In this section, we review existing robot motion retargeting methods, broadly categorizing them into three approaches: Configuration Space Retargeting (CSR), Task Space Retargeting (TSR), and Canonical Rig Retargeting (CRR). CSR methods directly map human joint angles to robot joints but typically require manual tuning for each robot's kinematic structure. TSR methods, on the other hand, focus on matching end-effector trajectories, resulting in more visually natural motions but necessitating careful handling of inverse kinematics and singularities. Recently, CRR methods have emerged, utilizing an intermediate canonical representation to unify diverse human skeleton inputs. However, existing CRR approaches often discard significant portions of input data due to feasibility constraints, limiting motion diversity. In contrast, our work explicitly integrates feasibility corrections and trajectory refinements within the retargeting pipeline, enabling robust retargeting across a wider range of human motions.

a) Configuration Space Retargeting (CSR): In CSR methods, robot joint angles are derived directly by mapping human joint values to the target robot, taking into account each robot's specific kinematic structure. For example, Safonova et al. [10] constrained upper-body gestures on a Sarcos humanoid platform by optimizing joint configurations. Yamane et al. [11] addressed whole-body retargeting using geometric considerations, while Ayusawa and Yoshida [12] proposed a model-based approach that uses an iterative scheme to fit the human motion onto the robot joints. Although CSR methods can be computationally efficient and straightforward to implement, they often require careful hand-tuning of the mapping between human and robot joints.

b) Task Space Retargeting (TSR): TSR methods focus on reproducing the positional and/or orientational trajectories

of key end-effectors (hands, feet, head) in the Cartesian space. By emphasizing global positions rather than joint angles, TSR approaches often preserve the look of the motion more effectively. Dariush et al. [13] introduced a markerless TSR-based retargeting with the Honda ASIMO, whereas Bin Hammam et al. [14] ensured kinodynamic consistency by enforcing task-level constraints. Darvish et al. [15] proposed a teleoperation framework using inverse kinematics on the iCub robot's end-effectors. TSR approaches can generate more human-like postures, but must still manage inverse kinematics solutions, possible singularities, and collisions in a separate step.

c) Canonical Rig Retargeting (CRR): CRR approaches unify diverse human skeleton sources into a single standardized rig before transferring motions to robots. Typically, raw human motions—often noisy or incomplete—are normalized onto this canonical representation, as shown by He et al. [16] and Liu et al. [17]. However, existing CRR methods frequently discard significant amounts of data during feasibility filtering, limiting motion diversity. Additionally, these methods typically separate retargeting from trajectory refinement, relying instead on external RL or model-based controllers to handle physical constraints. Such separation often causes mismatches between retargeted motions and controller assumptions, necessitating additional tuning.

In contrast, our approach explicitly performs feasibility corrections (e.g., foot contact, collision-free posture) and trajectory adjustments during the retargeting process itself, allowing a broader range of input motions to be converted into stable, collision-free robot trajectories.

Table I summarizes how our proposed method integrates and extends the key advantages of previous retargeting approaches. Specifically, our method explicitly addresses collision avoidance, leverages the benefits of refining human motions through a canonical rig representation, and incorporates trajectory-level adjustments directly into the retargeting pipeline.

TABLE I: Comparison of Motion Retargeting Approaches

Approach	Position-based Targets	Collision Awareness	Canonical Rig	Trajectory Adjustment
CSR [10]–[12]	×	✓	×	×
TSR [13]–[15]	✓	✓	×	×
CRR [16], [17]	✓	×	✓	×
Ours	✓	✓	✓	✓

III. PROPOSED METHOD

In this section, we propose a robust motion retargeting framework designed to generate feasible and expressive robot motions from diverse and often noisy human input data. Our approach first unifies diverse source skeletons into a standardized canonical representation (common-rig) using an optimization-based rigging process. Subsequently, we retarget this refined motion onto various robot platforms using direction vectors, explicitly addressing collisions, joint constraints, and stability. Finally, trajectory-level adjustments ensure the robot's end-effector motions closely follow the

source demonstrations. The overall motion retargeting process is summarized in Figure 2.

A. Problem Formulation

We aim to generate feasible, expressive robot motions across various kinematic configurations from source data. Our approach addresses noisy and inconsistent input data while ensuring the resulting robot motions adhere to kinematic constraints and maintain physical stability.

1) Notations:

- $\xi^{\mathcal{K}}$: Kinematic structure of a skeleton ($\mathcal{K} \in \{\text{mocap}, \text{rig}, \text{robot}\}$).
- $T_{1:L}^{\mathcal{K}} \in \text{SE}(3)^{L \times N}$: Homogeneous transformation matrices of skeleton joints over L time steps and N joints.
- $q_{1:L}^{\mathcal{K}} \in \mathbb{R}^{L \times N}$: Joint angles of the skeleton over L time steps and N joints.
- $\text{FK}(\xi^{\mathcal{K}}, q_{1:L}^{\mathcal{K}}) \mapsto T_{1:L}^{\mathcal{K}}$: Forward kinematics mapping joint angles to transformation matrices.

2) *Motion Retargeting Formulation*: We formalize our robust motion retargeting problem as follows:

$$\begin{aligned} \min_{q_{1:L}^{\text{robot}}} \quad & d(\hat{T}_{1:L}^{\text{robot}}, \text{FK}(\xi^{\text{robot}}, q_{1:L}^{\text{robot}})) \\ \text{s.t.} \quad & q_{1:L}^{\text{rig}} = \text{CR}(\xi^{\text{rig}}, \xi^{\text{mocap}}, T_{1:L}^{\text{mocap}}) \\ & T_{1:L}^{\text{robot}} = \text{RT}(\xi^{\text{robot}}, \xi^{\text{rig}}, q_{1:L}^{\text{rig}}) \\ & \hat{T}_{1:L}^{\text{robot}} = \text{FT}(T_{1:L}^{\text{robot}}, T_{1:L}^{\text{mocap}}) \\ & g(\xi^{\text{robot}}, q_{1:L}^{\text{robot}}) \geq 0 \end{aligned} \quad (1)$$

where:

- $\text{CR}(\cdot)$: Common-rigging step that unifies and refines source motion data into a standard auxiliary rig, addressing feasibility constraints.
- $\text{RT}(\cdot)$: Retargeting step that uses direction vectors from the common-rig to compute $T_{1:L}^{\text{robot}}$.
- $\text{FT}(\cdot)$: Fine-tuning step in which the end-effectors' trajectories are adjusted to follow the source motion in task space closely.
- $g(\cdot)$: Inequality constraints enforcing motion trajectory smoothness, self-collision avoidance, and COM stability for the robot.

B. Common-rigging

Motion data from various sources often have inconsistent skeleton structures and noisy or infeasible poses. We address this in two steps: *Pre-rigging* and *Post-rigging*.

1) *Pre-rigging*: In pre-rigging, we unify different skeletons into a single human-like common-rig ξ^{rig} , equipped with a standard joint hierarchy and physical properties (volume, mass). We define a set of joints of interest (JOI) [18] and solve an inverse kinematics problem to match the source JOI positions/orientations while enforcing a link-length constraint that preserves each source skeleton's proportions:

$$\min_{q_{1:L}^{\text{rig}}} d(T_{1:L}^{\text{mocap}}, \text{FK}(\xi^{\text{rig}}, q_{1:L}^{\text{rig}})) \quad \text{s.t. } \text{LL}(\xi^{\text{rig}}, \xi^{\text{mocap}})$$

Here, $\text{LL}(\cdot)$ ensures the common-rig's link lengths align with the source skeleton. We also adjust ankle orientations to keep the feet approximately parallel to the ground:

$$\min_{\hat{R}_{\text{ankle}}^{\text{rig}}} \|\hat{R}_{\text{ankle}}^{\text{rig}} \mathbf{e}_z^{\text{rig}} - \mathbf{e}_z\|_2$$

Incorporating physical properties such as mass and volume enables the rig to handle self-collisions and detect infeasible poses more effectively.

2) *Post-rigging*: Post-rigging refines lower-body motion to ensure physical feasibility by enforcing ground-contact constraints:

$$h(\xi^{\text{rig}}, q_{1:L}^{\text{rig}}) = 0,$$

which guarantee the feet remain fixed to the ground. We keep the torso upright to prevent the upper and lower body from rotating independently. Inequality constraints $g(\cdot)$ are applied to ensure stable and smooth motions. Specifically, we assume feet orientation parallel to the ground, constrain the COM projection within the convex support polygon formed by the feet, and prevent self-collisions by modeling body segments within capsule boundaries. Additionally, $g(\cdot)$ includes a smoothing constraint based on trajectory smoothing [19]. The result is a unified, feasible motion compatible with a wide range of robots.

C. Robot Motion Retargeting

After obtaining the refined common-rig motion, we retarget it onto each robot's specific kinematic structure as shown in Figure 3. This involves three main steps.

1) *Direction vector-based robot target pose*: We define JOI for the target robot, then extract direction vectors between relevant common-rig joints. These vectors are scaled by the robot's link lengths to compute the robot's target positions:

$$\mathbf{p}_j^{\text{robot}} = \mathbf{p}_i^{\text{robot}} + \mathbf{l}^{\text{robot}} \cdot \mathbf{v}^{\text{rig}},$$

where $\mathbf{p}_i^{\text{robot}}$ is the parent joint's position, $\mathbf{l}^{\text{robot}}$ is the robot's link length, and $\mathbf{v}^{\text{rig}}$ is the unit direction vector between corresponding joints on the common-rig. This approach accommodates differences in skeleton geometry while preserving the original motion's directional structure.

2) *Robot motion trajectory adjustment*: In many tasks, especially those requiring precise or expressive end-effector motions, we refine $T_{1:L}^{\text{robot}}$ in task space so that the robot's end-effectors more closely follow the source motion's trajectory. Concretely, let T_l^{mocap} represent the end-effector pose (e.g., position or homogeneous transform) from the source motion at time step l , and let T_l^{robot} be the preliminary retargeted pose for the robot. We define an affine mapping $f_\theta(\cdot)$, parameterized by θ (which may include scaling, rotation, and translation terms), to transform the source pose into a final, fine-tuned trajectory:

$$\hat{T}_{1:L}^{\text{robot}} = f_\theta(T_{1:L}^{\text{mocap}}), \quad \theta = \arg \min_{\theta} \sum_{l=1}^L \|f_\theta(T_l^{\text{mocap}}) - T_l^{\text{robot}}\|.$$

Once we obtain $\hat{T}_{1:L}^{\text{robot}}$, we solve an inverse kinematics problem to find $q_{1:L}^{\text{robot}}$ (the robot's joint angles over the motion

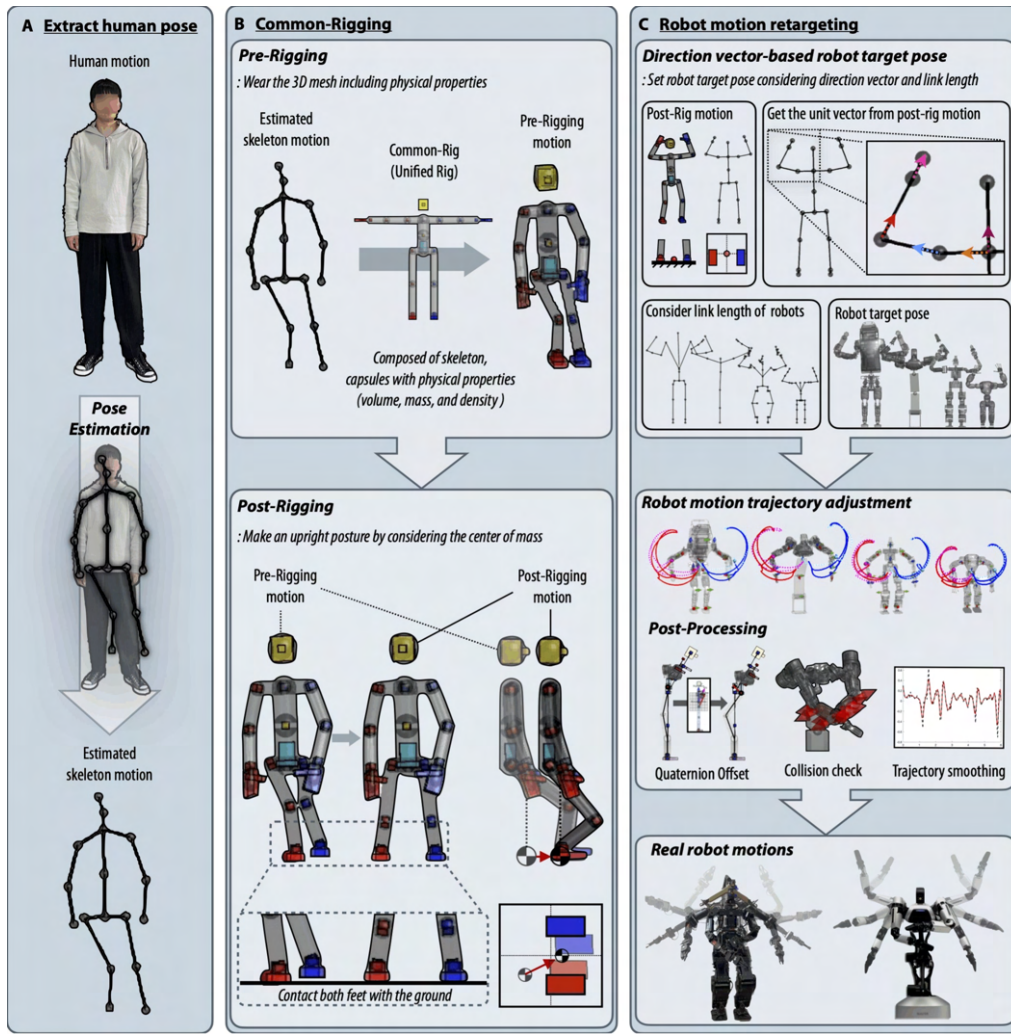


Fig. 2: Overview of proposed method. (A) Extracting human skeleton motion from motion capture systems or pose estimations. (B) Common-rigging process, which includes optimization-based methods called pre-rigging and post-rigging, aims to unify different human skeleton structures into a single predefined rig, referred to as a common-rig. (C) Robot motion retargeting consists of Direction vector-based robot target pose, Robot motion trajectory adjustment, and Post-processing. The process generates feasible real robot motions that preserve the features of the source human motion.

sequence) that minimize $\|\hat{\mathbf{t}}_{1:L}^{\text{robot}} - \text{FK}(\xi^{\text{robot}}, q_{1:L}^{\text{robot}})\|$ while satisfying inequality constraints $g(\cdot)$. This yields a feasible trajectory that remains similar to the end-effector movements in the source data. Figure 4 illustrates a comparison between the target trajectory and the robot’s end-effector trajectory.

3) *Post-processing*: Finally, we account for any remaining orientation mismatches between the common-rig and the robot geometry. In particular, the robot’s zero-pose joint orientations can differ from the rig’s zero-pose, which can introduce unwanted twists or shifts in the retargeted motion. To correct this, we compute a quaternion-based offset Q that rotates the rig’s direction vectors to match the robot’s baseline orientation. Specifically, if $\mathbf{v}_{ij}^{\text{rig}}$ and $\mathbf{v}_{ij}^{\text{robot}}$ denote the direction vectors from joint i to j in the rig and robot zero-poses, respectively, then Q is chosen such that

$$Q \cdot \mathbf{v}_{ij}^{\text{rig}} \approx \mathbf{v}_{ij}^{\text{robot}}.$$

Applying Q to the rig’s direction vectors ensures consistent relative poses once retargeted. We then apply a final smooth-

ing pass if necessary, making the retargeted motion feasible, collision-free, and faithful to the source motion’s style.

4) *Optimization Implementation Details*: We decomposed the overall trajectory optimization problem in Equation (1) into numerically solvable subproblems, facilitating efficient handling of nonlinear constraints and complex objectives. We employed numerical solvers, specifically gradient-based methods with line search, for inverse kinematics computations. MuJoCo was extensively utilized for accurate collision detection, Jacobian computations, and ensuring physical feasibility throughout the retargeting process. Solver tolerance thresholds were typically set to 10^{-3} to ensure numerical convergence. Using a standard consumer-grade laptop (MacBook Pro with an M3 processor), our pipeline typically generates a 10-second motion trajectory for a humanoid robot with approximately 25 degrees of freedom in about 1–2 minutes.

TABLE II: Performance evaluation (PA-MPJPE; lower is better) under clean inputs for 24 motions and 4 robot models.

No.	Motion Type	Atlas			COMAN			THORMANG			AMBIDEX		
		Ours	CSR [12]	TSR [15]	Ours	CSR [12]	TSR [15]	Ours	CSR [12]	TSR [15]	Ours	CSR [12]	TSR [15]
1	CMU Mocap - Wash windows	0.301	0.347	0.643	0.182	0.175	0.775	0.257	0.260	0.369	0.147	0.155	0.495
2	CMU Mocap - Jumping jacks	0.231	0.275	0.566	0.200	0.234	0.676	0.226	0.243	0.400	0.271	0.305	0.528
3	CMU Mocap - Direct traffic	0.437	0.510	0.622	0.247	0.297	0.698	0.325	0.348	0.579	0.242	0.285	0.712
4	CMU Mocap - Lean forward	0.265	0.361	0.774	0.336	0.396	0.668	0.346	0.350	0.424	0.247	0.272	0.376
5	CMU Mocap - Wait for bus	0.352	0.418	0.401	0.182	0.208	0.614	0.263	0.272	0.406	0.260	0.296	0.573
6	CMU Mocap - Penguin	0.221	0.233	0.914	0.215	0.234	0.810	0.416	0.426	0.589	0.234	0.234	0.859
7	CMU Mocap - Superhero	0.152	0.175	0.760	0.257	0.269	0.667	0.382	0.394	0.730	0.220	0.235	0.797
8	CMU Mocap - Lambada dance	0.306	0.427	1.000	0.227	0.275	0.836	0.390	0.431	0.686	0.258	0.320	0.793
9	CMU Mocap - Eating dinner	0.432	0.482	0.649	0.366	0.427	0.755	0.304	0.317	0.625	0.360	0.379	0.552
10	CMU Mocap - Playing violin	0.325	0.351	0.533	0.178	0.189	0.467	0.344	0.355	0.472	0.210	0.220	0.552
11	CMU Mocap - Driving	0.364	0.401	0.593	0.199	0.210	0.926	0.351	0.361	0.573	0.291	0.304	0.561
12	CMU Mocap - Stretch and yawn	0.340	0.347	0.431	0.314	0.313	0.361	0.255	0.259	0.375	0.325	0.332	0.456
13	Emotion Mocap - Surprise	0.177	0.232	0.479	0.150	0.231	0.441	0.123	0.173	0.425	0.168	0.190	0.594
14	Emotion Mocap - Sad	0.133	0.135	0.354	0.108	0.164	0.578	0.116	0.173	0.397	0.149	0.232	0.460
15	Emotion Mocap - Angry1	0.224	0.291	0.545	0.139	0.184	0.500	0.184	0.295	0.463	0.199	0.272	0.688
16	Emotion Mocap - Angry2	0.176	0.195	0.472	0.107	0.154	0.544	0.128	0.167	0.403	0.104	0.255	0.402
17	Emotion Mocap - Angry3	0.228	0.243	0.626	0.161	0.263	0.654	0.147	0.196	0.365	0.193	0.247	0.671
18	Emotion Mocap - Angry4	0.138	0.197	0.462	0.131	0.197	0.436	0.110	0.155	0.312	0.148	0.197	0.585
19	Emotion Mocap - Angry5	0.169	0.186	0.515	0.127	0.174	0.469	0.167	0.194	0.396	0.145	0.167	0.573
20	Emotion Mocap - Disgust1	0.146	0.188	0.497	0.160	0.171	0.581	0.115	0.145	0.416	0.149	0.183	0.583
21	Emotion Mocap - Disgust2	0.174	0.199	0.542	0.164	0.191	0.554	0.173	0.244	0.502	0.167	0.249	0.665
22	Emotion Mocap - Fearful1	0.171	0.173	0.596	0.138	0.174	0.508	0.143	0.168	0.435	0.111	0.142	0.562
23	Emotion Mocap - Fearful2	0.125	0.153	0.429	0.095	0.155	0.389	0.076	0.114	0.371	0.106	0.150	0.487
24	Emotion Mocap - Happy	0.168	0.183	0.421	0.122	0.203	0.469	0.171	0.197	0.362	0.162	0.180	0.360

TABLE III: Robustness evaluation (PA-MPJPE; lower is better) under noisy inputs for 12 motions and 4 robot models.

No.	Motion Type	Atlas			COMAN			THORMANG			AMBIDEX		
		Ours	CSR [12]	TSR [15]	Ours	CSR [12]	TSR [15]	Ours	CSR [12]	TSR [15]	Ours	CSR [12]	TSR [15]
1	CMU Mocap - Unscrew bottlecap	0.146	0.266	0.430	0.149	0.233	0.485	0.209	0.234	0.223	0.122	0.249	0.362
2	CMU Mocap - Chicken dance	0.063	0.125	0.551	0.109	0.151	0.602	0.110	0.138	0.266	0.345	0.554	0.518
3	CMU Mocap - Bear	0.057	0.089	0.278	0.059	0.104	0.562	0.090	0.223	0.623	0.120	0.177	0.361
4	CMU Mocap - Crying	0.208	0.409	0.577	0.442	0.508	0.654	0.348	0.496	0.562	0.324	0.572	0.463
5	CMU Mocap - Scared	0.239	0.249	0.648	0.265	0.253	0.697	0.235	0.240	0.603	0.410	0.463	0.537
6	CMU Mocap - Upset	0.115	0.151	0.376	0.112	0.096	0.321	0.112	0.200	0.143	0.164	0.169	0.359
7	CMU Mocap - Chugging	0.172	0.302	0.361	0.201	0.228	0.396	0.192	0.212	0.367	0.183	0.237	0.271
8	CMU Mocap - Making coffee	0.156	0.230	0.522	0.150	0.173	0.562	0.134	0.160	0.222	0.151	0.207	0.541
9	CMU Mocap - Violin	0.077	0.131	0.566	0.091	0.120	0.309	0.119	0.109	0.253	0.066	0.107	0.436
10	CMU Mocap - Waving	0.050	0.107	0.346	0.058	0.077	0.355	0.050	0.079	0.274	0.048	0.089	0.398
11	CMU Mocap - Macarena dance	0.123	0.138	0.488	0.095	0.082	0.358	0.093	0.084	0.249	0.099	0.118	0.801
12	CMU Mocap - Sun salutation	0.045	0.047	0.268	0.065	0.072	0.258	0.039	0.240	0.205	0.039	0.257	0.361

IV. EXPERIMENT

In this section, we validate our approach on (i) extensive simulations with 12 different humanoid robots, (ii) challenging motion capture data and noisy pose estimations, and (iii) real-world demonstrations on three hardware platforms: AMBIDEX, THORMANG, and JF2. Our experiments focus on foot-fixed scenarios (i.e., no stepping), showcasing how the proposed pipeline reliably retargets expressive upper-body motions while maintaining full-foot contact and satisfying kinematic, collision, and balance constraints.

A. Simulated Robots and Data Sources

Robotic Platforms. We demonstrate our pipeline on 12 simulated humanoids, including Atlas, COMAN, H1, JAXON, Valkyrie, Talos, ErgoCub, iCub, Darco3, and three robots available in hardware form: AMBIDEX, THORMANG, and JF2 (Figure 5). Each robot features unique link lengths, joint configurations, and mass distributions; this diversity highlights the scalability of our *common-rigging* and *retargeting* steps.

Motion Data. We collected a wide range of source motions, categorized as follows:

- *High-fidelity MoCap data* (e.g., CMU MoCap [5], dynamic emotional expressions [6]).
- *Noisy pose-estimation data* from RGB videos [7]–[9], including casual phone recordings with imperfect 3D

pose reconstructions.

These inputs encompass everyday tasks (waving, dancing, opening a bottle), artistic performances (violin playing, contemporary dance), and expressive gestures (angry, sad, surprised). Many of them contain significant torso motion, bent knees, and wide arm swings—challenging poses for robots in a foot-fixed stance.

B. Qualitative Results

Figure 5 illustrates our method’s ability to retarget various human motions across multiple robot platforms. Even with the lower body fixed on the ground, robots produce smooth, dynamic upper-body gestures, for instance, dancing or playing violin—closely resembling the source motion. By carefully enforcing constraints in the *post-rigging* step, our approach ensures that the feet remain flat on the ground, the center of mass stays within the support polygon, and collisions are avoided.

C. Quantitative Evaluation and Comparison

To quantify motion expressiveness, we compare the upper-body trajectories (shoulders, elbows, wrists) of the retargeted motion to those of the original human motion. We use the commonly adopted *Procrustes Aligned Mean Per Joint Position Error (PA-MPJPE)* [20]: lower values indicate better preservation of the original movement pattern.

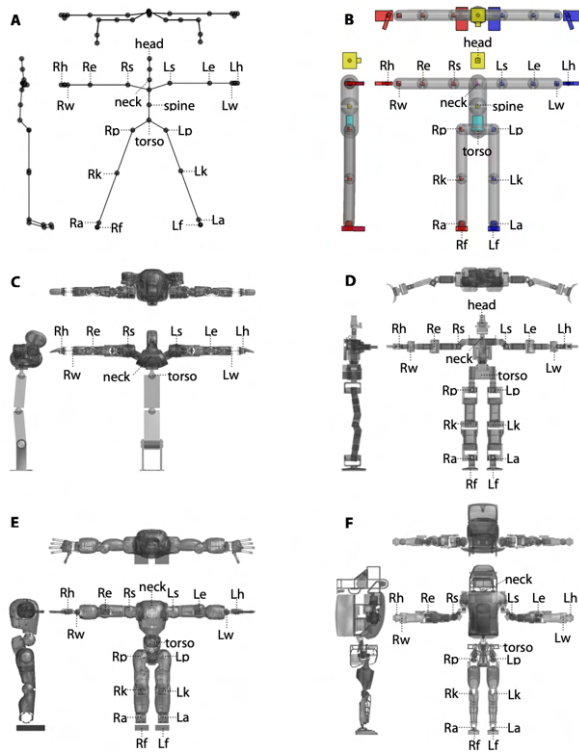


Fig. 3: Kinematic structures and joints of interest (JOI) information. (A) Estimated human skeleton. (B) Common-rig. (C) AMBIDEX (D) THORMANG (E) COMAN (F) ATLAS

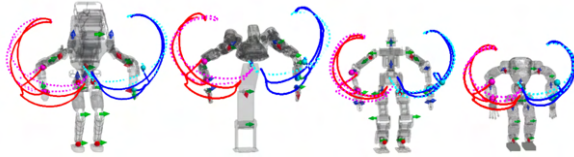


Fig. 4: Comparison between the target trajectory from human motion (dotted) and the adjusted end-effector trajectories for various robots (solid).

We benchmark our pipeline to two established motion-retargeting baselines:

- 1) *Configuration Space Retargeting (CSR)* [12] – directly maps joint angles from human skeletons onto a robot’s configuration space with minimal IK.
- 2) *Task Space Retargeting (TSR)* [15] – focuses on matching end-effector positions/orientations but without the full collision/balance refinement of our pipeline.

Motion retargeting methods are evaluated on both *clean* motions (from precise MoCap) and *noisy* motions (with artificially injected Gaussian noise or raw 3D pose estimations). Tables II and III present results for Atlas, COMAN, THORMANG, and AMBIDEX across various test scenarios. Our method consistently outperforms *CSR* [12] and *TSR* [15] in preserving the original motion style (i.e., achieving lower *PA-MPJPE*). Notably, when large torso or limb inaccuracies are present, our two-step *common-rigging* (Figure 6) robustly corrects foot misalignments and self-collisions while maintaining expressive upper-body trajectories.

Existing *CRR* approaches [16], [17] typically handle col-

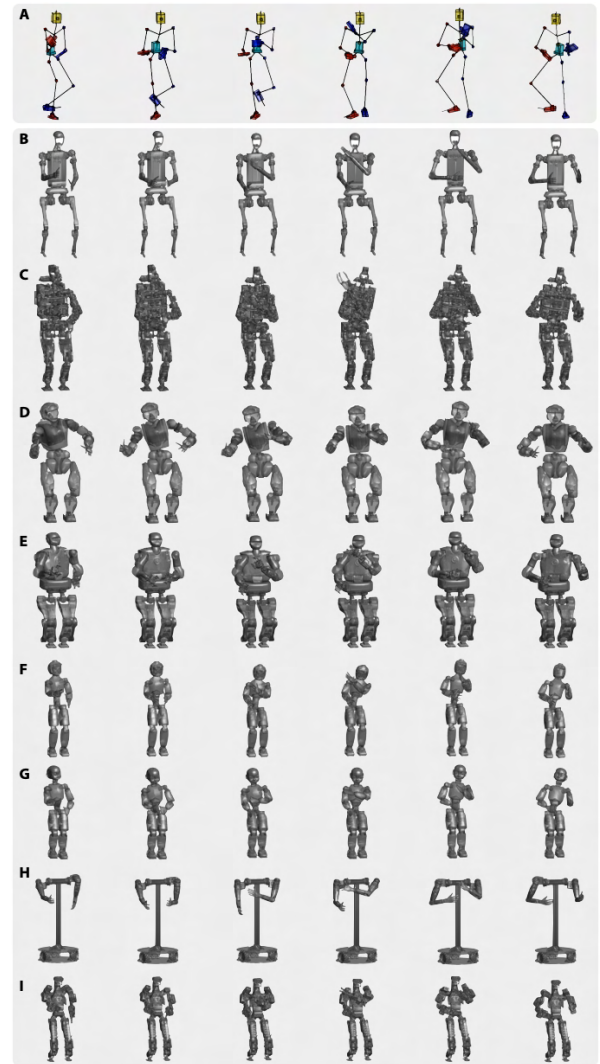


Fig. 5: Motion retargeting to multiple robots: (A) Human motion and retargeted motions for (B) H1, (C) Jaxon, (D) Valkyrie, (E) Talos, (F) ErgoCub, (G) iCub, (H) JF2, (I) Darco3.

lisions and infeasible poses by either discarding significant portions of the input data or relying on external controllers for trajectory refinement after retargeting. This separation of retargeting and refinement stages makes direct quantitative comparisons challenging, as the final motion quality in *CRR* strongly depends on additional filtering or controller-specific tuning. In contrast, our integrated pipeline explicitly incorporates collision checks, balance constraints, and trajectory adjustments during the retargeting itself, enabling robust handling of noisy data without significant data loss. For these reasons, we focus our quantitative evaluations specifically on the *CSR* and *TSR* approaches, where direct comparisons are more straightforward.

D. Real Robot Demonstrations

Finally, we validate the feasibility of our approach on three real humanoids: AMBIDEX, THORMANG, and JF2. Figure 7 demonstrates retargeted motions for tasks like *violin playing* and *penguin walk* from the CMU MoCap dataset;

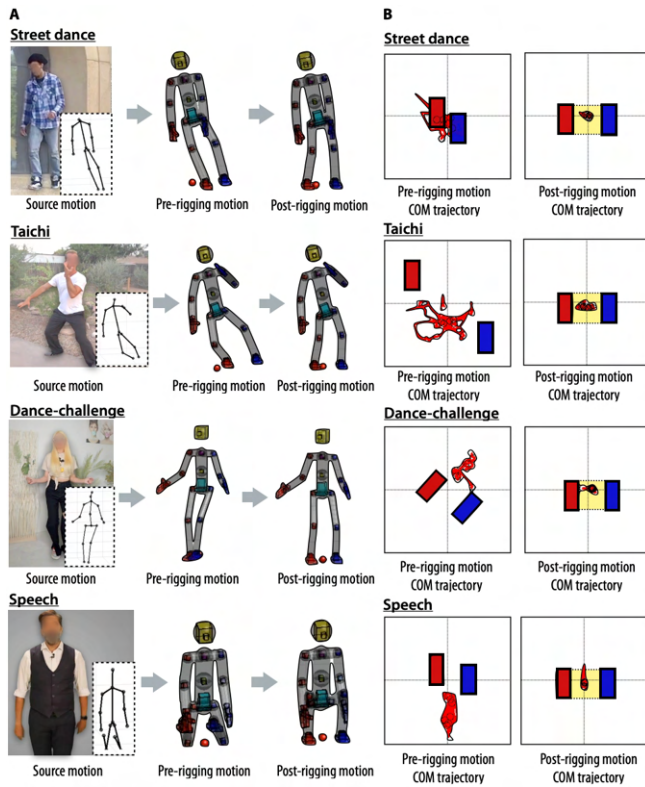


Fig. 6: (A) Refining noisy torso/foot poses to an upright posture with feet on the ground. (B) Center-of-mass trajectory stays inside the contact polygon when using our post-rigging.

Figure 8 shows robot executions from phone-recorded videos (using 3D pose estimation). Despite the inherent noise in the video-based data, post-rigging enforces stable foot contact and self-collision avoidance, enabling the real robot to effectively reproduce complex upper-body gestures.

In all hardware trials, we observe:

- *Smoothness and Expressiveness*: Upper-body motions capture arm swings and torso bends closely matching the source video or MoCap.
- *Physical Feasibility*: Robots remain fully balanced with grounded feet, and no self-collision or excessive joint torque is observed.
- *Minimal Tuning Effort*: The same pipeline parameters (e.g., optimization thresholds) transfer across different robots and motions, highlighting the method’s robustness.

E. Discussion and Limitations

Overall, our results show that refining human input motion into a canonical rig and then applying trajectory-level adjustments with strict feasibility constraints yields realizable and highly expressive robot movements in foot-fixed scenarios. By carefully enforcing kinematic constraints, our approach maintains key stylistic elements—such as torso and limb expressiveness while keeping both robot feet in contact with the ground. Even though foot contact is fixed, subtle lower-body adjustments, including controlled knee bends and torso movements, significantly enhance the overall expressiveness of motions. Figure 9 demonstrates how expressive gestures

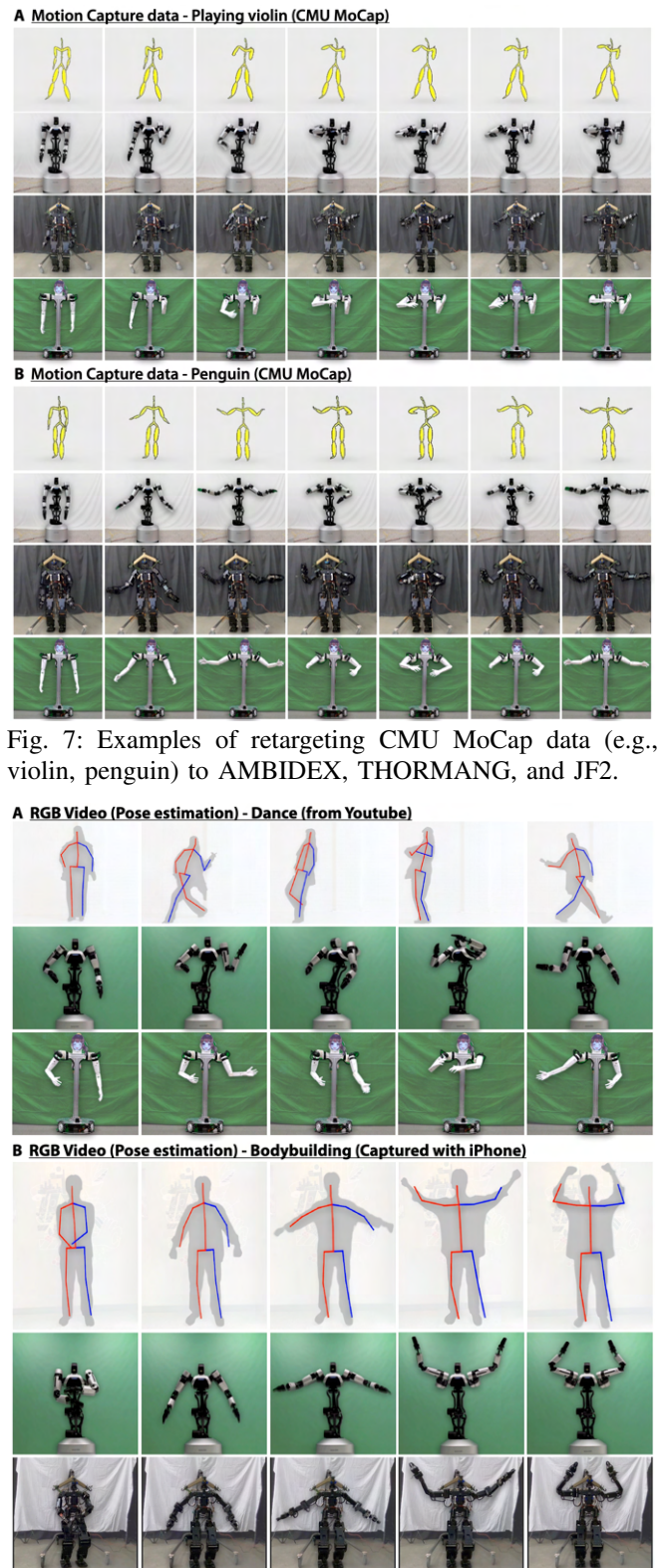


Fig. 7: Examples of retargeting CMU MoCap data (e.g., violin, penguin) to AMBIDEX, THORMANG, and JF2.

Fig. 8: Retargeting human videos to AMBIDEX, THORMANG, and JF2 using pose estimation.

and emotional nuances from a “Happy” motion can still be vividly conveyed through torso inclination and knee movements, even when both feet remain fixed to the ground.

Although we have not explicitly addressed dynamic stepping or locomotion, our method achieves stable and ex-

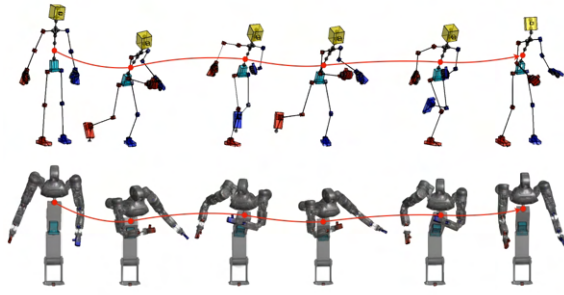


Fig. 9: Robot retargeting of “Happy” motion. Despite ground-fixed feet, torso and knee movements effectively preserve the original motion characteristics.

pressive upper-body movements, enhanced by subtle lower-body adjustments. Such features are particularly valuable in applications like entertainment, telepresence, and interactive displays, where robots generally remain stationary. Future work will extend our approach to whole-body humanoid motion control using reinforcement-learning-based controllers [21] to enable dynamic foot-lifting behaviors. We anticipate that integrating our robust motion retargeting framework as an initial step in humanoid whole-body control could significantly improve task success rates and reduce the complexity typically required in RL-based fine-tuning.

V. CONCLUSION

We presented a robust and automated approach for retargeting expressive human motions onto diverse humanoid robots by first refining noisy input data through a unified canonical rig and subsequently enforcing strict kinematic constraints, including stable foot contact, collision avoidance, and center-of-mass alignment. Extensive experiments demonstrated our pipeline’s efficacy across both simulated and physical humanoids, reliably preserving upper-body expressiveness despite foot-fixed constraints. Future work will focus on integrating our framework with reinforcement-learning-based whole-body controllers to enable dynamic stepping behaviors, potentially enhancing performance and generalization in complex humanoid tasks.

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